Mikhail Razakov

Compiler and Runtime Engineer

misha13022008@gmail.com
t.me/Micodiy · github.com/Misha1302

I build UniversalToolchain/Wist2, a modular .NET platform with multiple IRs, a reference interpreter, a CIL/DynamicMethod backend and a verifier-gated AIR → SSA → AIR route. Additional depth includes conservative program analysis, C++23 and x86 code generation.

AIR → SSA → AIR Optimizations and structural verifiers	**1,358 / 1,358 tests** Latest full run; build succeeded	**1st of 49** PS-form Analyzer, 104/104

EXPERIENCE

2024 - PRESENT

UniversalToolchain / Wist2 - creator and primary developer

- Designed Source -> AST -> Bytecode -> AIR -> execution, a reference interpreter and a CIL/DynamicMethod backend.
- Implemented opt-in AIR → SSA → AIR with constant folding, SCCP-lite, branch folding, unreachable cleanup and DCE.
- Added capability contracts, structural verifiers and interpreter/CIL parity checks for supported semantics.

JUL - AUG 2026

MCST - compiler engineering intern

- C++23 graph-analysis components under CMake, warnings-as-errors, ASan/UBSan, clang-tidy and explicit includes.
- RPO with cycle detection, Dijkstra, Tarjan SCC and Dinic over a shared graph model and reproducible I/O tests.

SELECTED PROJECTS

PS-form Memory Dependence Analyzer

C17 / PROGRAM ANALYSIS

Conservative inter-iteration memory-dependence analysis. Ranked 1st of 49 with the only 5.0/5.0 and 104/104 result; exact oracle plus randomized and metamorphic tests.

NASM IA-32 and x86-64 codegen

NASM COURSE

NASM IA-32: cdecl, stack frames, libc, structures/addressing and x87. Separate x86-64 SysV emitter lab: liveness, live intervals, linear scan, iced-x86 and interpreter-vs-native validation.

CORE SKILLS

Compiler / IR
parsing, AST, Bytecode, AIR, CFG, SSA, CIL/DynamicMethod

Analysis / optimization
dominance, liveness, SCCP-lite, DCE, memory-dependence analysis

Correctness
structural verifiers, exact oracles, differential and metamorphic testing

Low-level
NASM IA-32, cdecl, stack/x87; x86-64 SysV codegen, objdump/disassembly

Languages / tools
C#/.NET, C++23, C17, Python; CMake, GitHub Actions, sanitizers, clang-tidy

EDUCATION AND AVAILABILITY

HSE University student. Moscow / remote. From September 2026: up to 20 hours per week.

TEACHING

Taught about 50 students in C/C++, Python and algorithms; authored a practical NASM IA-32 course.

RECOGNITION

HSE Olympiad prize-winner; 1st by total score in MEPhI Junior (2025 Engineering Sciences; 2026 IT); Baltic competition Grand Prize.